

Vandan Agrawal

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Software Engineer with production experience across full-stack development, RAG/LLM systems, and security principles. Builds and ships end-to-end, with a track record of measurable impact across startups and security research.

Education

University of Florida

December 2025

Master of Science in Computer Science | GPA: 3.66

Amity University

May 2024

Bachelor of Technology in Computer Science and Engineering

Coursework: Distributed OS Principles, Advanced Computer Networks, Programming Language Principles, Trustworthy Machine Learning, Malware Reverse Engineering, Intro to Cryptology, Cyber-Physical Systems Security

Skills

- **Languages:** Python, TypeScript, Java, C++, C, Kotlin, SQL, x64 assembly
- **Frameworks and Libraries:** Next.js, Flutter, Tkinter, Firebase, Groq Cloud, FastAPI
- **Developer Tools:** Google Cloud Platform, Claude, Git, AWS, Vertex AI
- **Security tools:** Ghidra, x64dbg, VMware, Wireshark, Splunk
- **Operating Systems:** Windows, Linux (Debian, Kali, Ubuntu), MacOS

Experience

Software Engineer, Skedaddle Enterprises

March 2026 – Present

- Scaled to 5000+ users by engineering a RAG-based AI system with Llama 3.3 and Groq API to process Firestore interest vectors, generating personalized member Context-Cards and activity suggestions.
- Migrated Swift/Flutter app to native Kotlin, unifying the Android stack and accelerating deployment cycles.
- Integrated Ticketmaster and Google Places APIs for location-aware, proximity-based event discovery.
- Built real-time RSVP/notification system with NoSQL Firestore listeners, supporting 10000+ concurrent users.
- Aligned API contracts across Android and Core infrastructure teams to maintain cross-platform logic parity.

Cybersecurity Analyst, Malware Reverse Engineering Lab

January 2025 – May 2025

- Reverse-engineered over 10 real-world malware samples analyzing direct system calls in x64 assembly to identify execution flow, persistence, C2, and IOCs, enabling rapid incident response against threats.
- Ran 60+ hours of static/dynamic analysis using Ghidra, Wireshark, and x64dbg in a multi-system sandbox.
- Reduced manual reverse-engineering time 90% by automating obfuscation-bypass with Python scripts.
- Dissected a variant of the Akira ransomware as a final project, documenting behavior and IOCs for a threat that has impacted over 250 small to medium-sized American organizations since March 2023.
- Authored YARA rules for malware family detection, strengthening endpoint security posture.

Software Developer Intern, SmartInternz

May 2023 – July 2023

- Deployed 20+ Salesforce objects/workflows enforcing secure, role-based access control.
- Built async Apex triggers/batch processes for high-concurrency data, improving responsiveness 15%.
- Automated compliance monitoring for 5000+ records via Shield Encryption, cutting manual review by 60%.
- Orchestrated 500+ weekly transactions via REST API integrations and Screen Flows.

Projects

Clinical Trial Immune Cell Analytics Dashboard - Python, SQLite, FastAPI

August 2026

- Designed normalized SQL schema for clinical trial dataset consisting of 3,500 subjects and 52,500 records.
- Enforced data integrity via composite unique constraints, preventing cross-project subject ID collisions.
- Built statistical analysis pipeline comparing responders vs. non-responders across immune cell populations.
- Authored production scaling plan for SQLite to Postgres migration and query precomputation as trial data grows.

Municipal Research & Analysis Engine - Google Cloud Platform (GCP), Claude Code, TypeScript, Python

June 2026

- Architected multilingual LLM search engine on GCP with Next.js/FastAPI, adding OAuth 2.0 and rate-limiting.
- Developed RAG pipeline with Vertex AI Search and Gemini 2.5, automating daily incremental ingestion from New York City legislative APIs to keep 8000+ municipal documents synchronized and searchable
- Added Google OAuth and Firestore-backed notebook for persistent bookmarking and search history.

Practical Defenses for Deployed AI Models - Python, PyTorch

December 2025

- Built a defense system that protects AI models from two real attacks: one that tricks them into wrong answers, another that leaks whether someone's data was used to train them.
- Raised accuracy under adversarial attack from 6.8% to 60.3% and cut membership inference advantage by about 95%, using inference-time defenses only, all without retraining the underlying model from scratch

Distributed Reddit Concurrency Engine and Simulator - Go (Golang)

December 2024

- Designed a lock-free, actor-based distributed system in Go (Proto.Actor) simulating a Reddit backend with concurrent user registration, subreddit management, post voting, karma calculation, and direct messaging.
- Modeled user activity via Zipf distribution to simulate non-uniform subreddit membership and posting.
- Benchmarked performance up to 100000 concurrent actors, profiling memory, and goroutines using Go pprof.